Buffer system
Acidosis & Alkalosis

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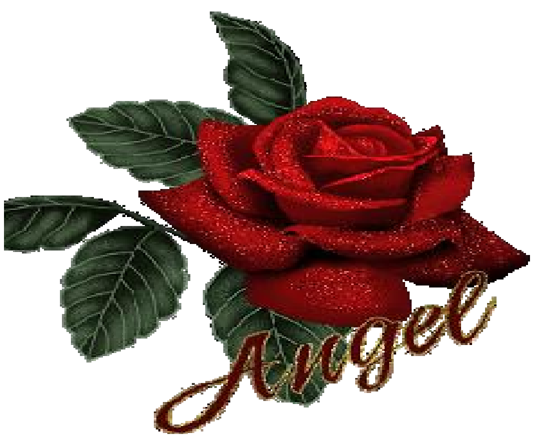


By the end of this lecture, the student will be able to:

A1. Define acids, base, pH.

A2. Point out the regulatory mechanisms of acid base homeostasis

A3. Describe acid base disorders and their types, causes, compensatory mechanisms.

A4. Interpret arterial blood gases report.

Acid Base Balance

- **Human diet is almost neutral but the metabolic processes in the body produce large amounts of acids like carbonic , Citric, sulfuric, phosphoric….etc.**
- **These products are transported to the lung (co_2) and kidney (sulfuric, phosphoric) via blood and ECF without changing arterial PH (7.35 - 7.45) and venous PH (7.32-7.38).**

<u>This balance is owing to 3 important regulatory mechanisms:</u>

1- Buffering capacity of blood, React very rapidly (less than a second)

2- Respiratory regulatory mechanisms. Reacts rapidly (seconds to minutes)

3- Renal regulatory mechanisms. Reacts slowly (minutes to hours)

Effects of pH changes

- The most general effect of pH changes are on enzyme function, also hormones and ion distribution are all affected by Hydrogen ion concentrations.

- Any small change in pH will produce great change in cell functions including enzyme activity, ion distribution…etc

Acids are sources of H+ ions

Acids can be defined as chemical substances that donate protons (H+)

Strong acids readily give protons (proton donors)

Physiologically important acids include:

- **Carbonic acid (H_2CO_3)**
- **Phosphoric acid (H_3PO_4)**
- **Pyruvic acid ($C_3H_4O_3$)**
- **Lactic acid ($C_3H_6O_3$)**

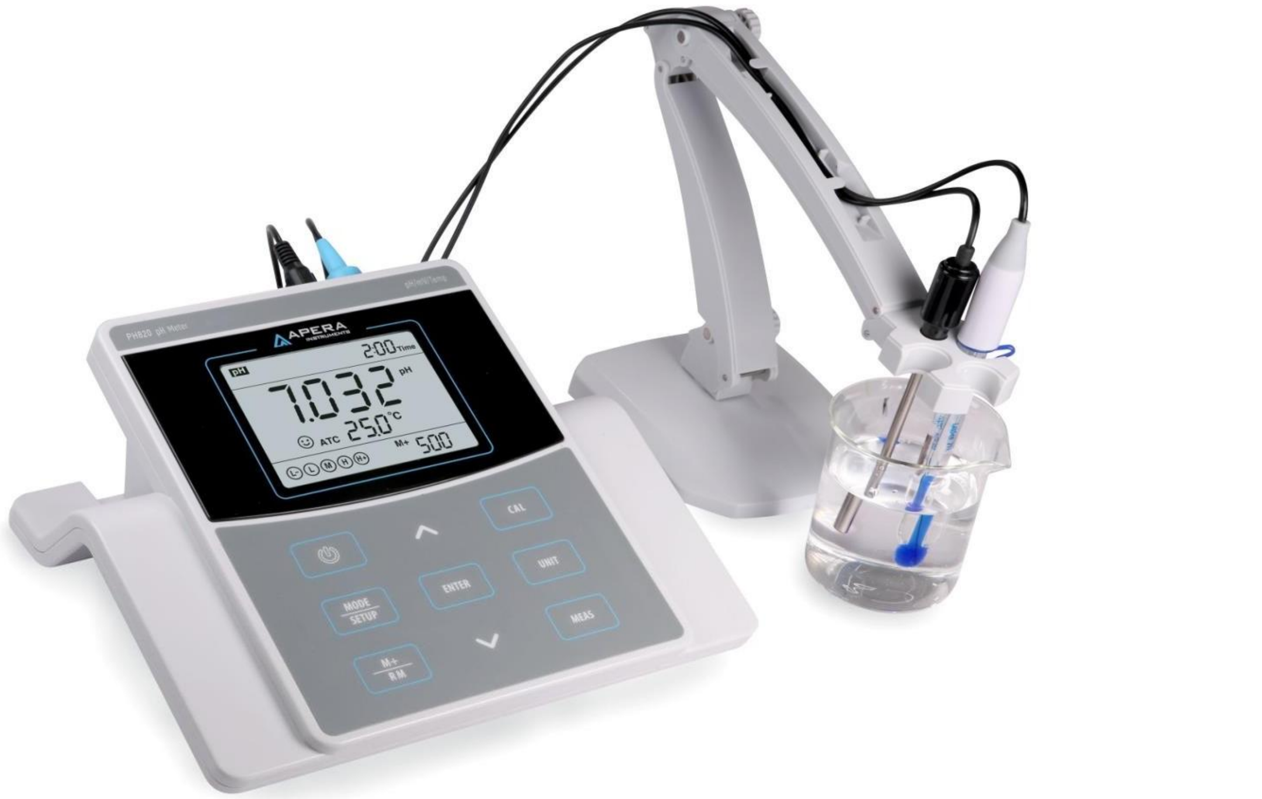
Bases

Bases can be defined as a substance that accept protons (H+)

Strong bases readily accept protons (proton acceptor)

Physiologically important bases include:

Bicarbonate (HCO_3^-) and Bi-phosphate (HPO_4^{2-})



pH Meter for measurement of pH of solutions

Normal Body pH

Normal arterial blood pH: 7.35 – 7.45

Below 7.35: Acidosis

Above 7.45: Alkalosis

Small changes can disrupt cell function

What is a Buffer?

- A system that resists pH change
- Composed of a weak acid and its conjugate base
- Neutralizes added acids or bases
- Acts rapidly to stabilize pH

Major Buffer Systems in the Body

- **Bicarbonate buffer system**
- **Phosphate buffer system**
- **Protein buffer system**
- **Hemoglobin buffer system**

Bicarbonate Buffer System

- **Most important extracellular buffer**
- **Components: Carbonic acid (H_2CO_3) and bicarbonate (HCO_3^-)**
- **Reaction: $CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$**
- **Regulated by lungs and kidneys**

Role of the Lungs

- **Control carbon dioxide (CO_2) levels**
- **CO_2 acts as an acid in the body**
- **Increased breathing removes CO_2 → raises pH**
- **Decreased breathing retains CO_2 → lowers pH**

Role of the Kidneys

- **Regulate bicarbonate (HCO_3^-) levels**
- **Excrete hydrogen ions (H^+) in urine**
- **Reabsorb or generate bicarbonate**
- **Provide long-term pH regulation**

Phosphate Buffer System

- Important in intracellular fluid
- Also active in kidney tubules
- Components: $H_2PO_4^-$ (acid) and HPO_4^{2-} (base)
- Helps regulate urinary pH

Protein Buffer System

- Proteins contain amino acids that can accept or donate H+
- Present in blood plasma and cells
- Important intracellular buffers
- Includes hemoglobin in red blood cells

A. Bicarbonate buffer system

Bicarbonate Buffer System HCO_3^- / H_2CO_3

Normal ratio of HCO_3^- to H_2CO_3 is 20:1

The chief buffer of blood

B. Phosphate buffer system

Phosphate buffer system HPO_4 / H_2PO_4

The chief buffer of urine.

C.Protein buffer system

Behaves as a buffer in both plasma and cells. Albumin is the most important in plasma. Hemoglobin is by far the most important protein buffer inside erythrocyte.

Abnormal Acid-Base Status

Acid base disturbances are traditionally classified into:

Metabolic acidosis

Metabolic alkalosis

Respiratory acidosis

Respiratory alkalosis

Normal values of Arterial Blood Gases (ABG)

PH 7.35 – 7.45

$PaCO_2$ (35 – 45) mmHg

HCO_3 (22 – 26) mEq /L

PaO_2 (80 – 100) mmHg

O_2 Saturation (95 % – 98)%

Acidosis

- **Acidosis** results from increasing hydrogen ion concentration of arterial blood plasma
- Generally acidosis occurs when *pH* of the blood decreases below 7.35
- Acidosis is either due to **Respiratory** or **Metabolic** causes

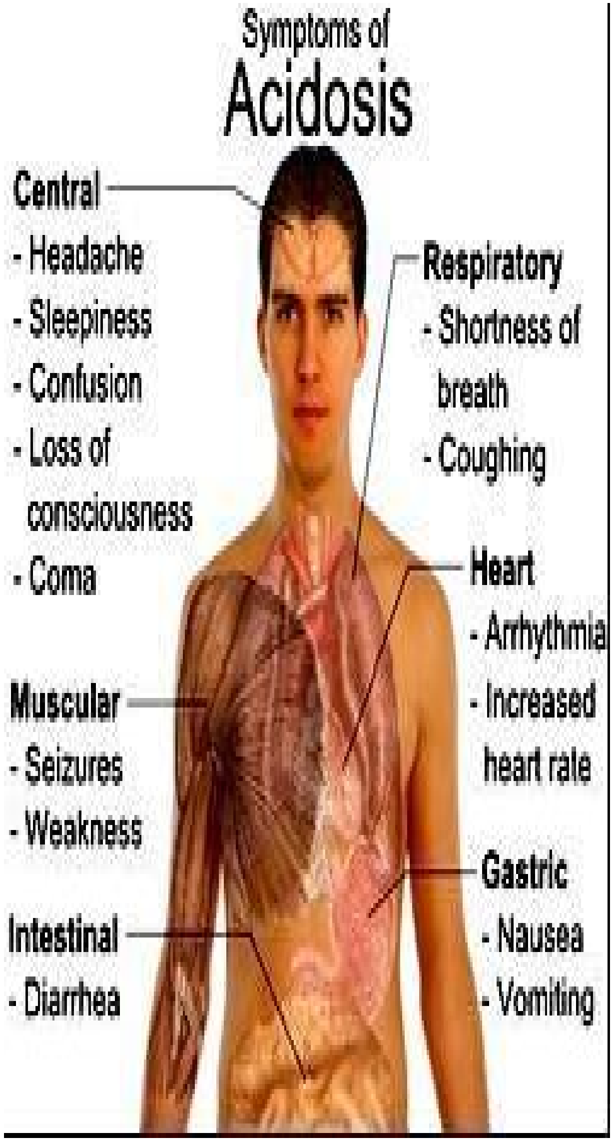
A. Respiratory acidosis

It results from any condition decreasing the rate of respiration = **hypoventilation**.

- In respiratory acidosis, the **CO_2 is increased** while the **bicarbonate** is either **normal** or **increased**

Causes of respiratory acidosis:

1. **Pulmonary** problems e.g.: **Pneumothorax**, **emphysema**, **chronic bronchitis**, **asthma**, severe **pneumonia**, and **aspiration**
2. **Head injuries**
3. **Drugs** (especially **anaesthetics** and **sedatives**)
4. **Brain tumors**



The response to acidosis as follows: By increasing renal reabsorption of Na HCO3 to restore the ratio. increase

urinary excreation of H+, NH4+ and CL- to help reabsortion of Na HCO3

B. Metabolic acidosis

□ Metabolic acidosis results from the increased production of metabolic acids

□ In metabolic acidosis, the CO2 is normal while the bicarbonate is decreased

Causes of metabolic acidosis:

1. **Renal acidosis** is due to renal failure in which, there is decrease in the ability of kidneys to excrete acid with an accumulation of urea and creatinine.
2. **Lactic acidosis** may occur from: **Hypoxemia** and **Hypoperfusion**
3. **Starvation and diabetic ketoacidosis** is due to the accumulation of ketoacids (ketosis).
4. **Severe or chronic diarrhea.**

The kidneys response to acidosis as follows:

□ Tubular cells **reabsorb more bicarbonate** from the tubular fluid to the blood

□ Collecting duct cells **excrete more hydrogen** and **generate more bicarbonate**, and **increase formation of the NH3**.

Alkalosis

- Alkalosis results from **reducing hydrogen ion concentration** of **arterial blood plasma.**
- Generally alkalosis occurs when *pH* of the blood exceeds 7.45
- Alkalosis is either due to **respiratory** or **metabolic** causes

A. Respiratory alkalosis

- It results from any condition increasing the rate of respiration = hyperventilation
- In respiratory alkalosis, the CO_2 is decreased
- There are two types of respiratory alkalosis: **acute** and **chronic**:

1. Acute respiratory alkalosis occurs rapidly and the person may lose consciousness

2. Chronic respiratory alkalosis is a more long-standing condition

Causes of respiratory alkalosis:

1. Excessive **mechanical ventilation**
2. **Psychiatric** causes: anxiety, hysteria and stress
3. **CNS causes**: stroke, meningitis
4. **Drug use**: aspirin, caffeine
5. **High altitude areas**, where the low oxygen stimulates increased ventilation
6. **Fever**
7. **Pregnancy**

The lungs response to alkalosis as follows:

- By decreasing breathing, expelling less CO_2

B. Metabolic Alkalosis

Causes of metabolic alkalosis

1. Prolonged vomiting, resulting in a loss of hydrochloric acid

2. Consumption of alkali

3. Endocrine disorders such as **Cushing's syndrome and associated with hypokalaemia.**

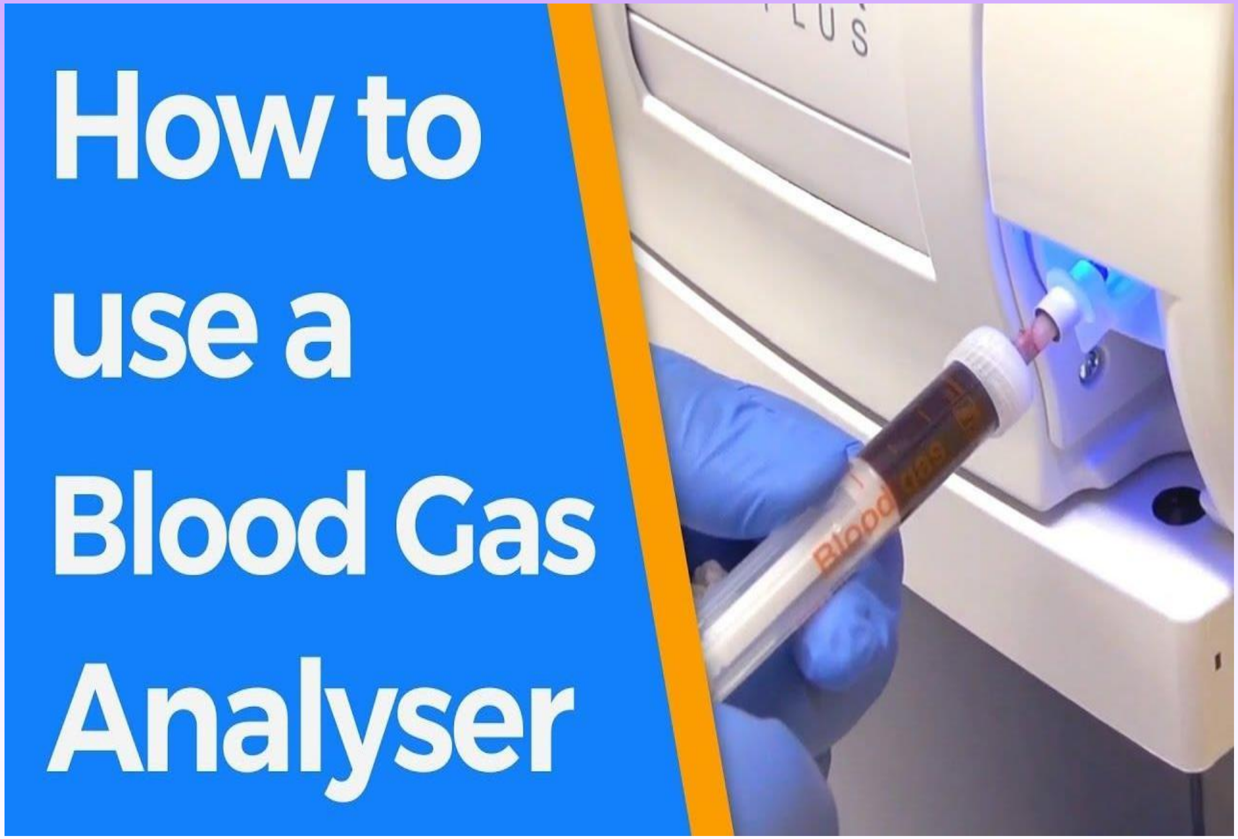
The kidneys response to alkalosis by:

1. Excreting more bicarbonate in urine
2. Decreasing hydrogen ion excretion from the tubular epithelial cells
3. Increasing glutamine synthesis

Interpreting ABGs

1. Look at the pH: is the primary problem acidosis (low) or alkalosis (high)
2. Check the CO_2 (respiratory indicator) :is it less than 35 (alkalosis) or more than 45 (acidosis)
3. Check the HCO_3 (metabolic indicator) is it less than 22 (acidosis) or more than 26 (alkalosis)
4. Which is primary disorder (Resp. or Metabolic)?

ABG apparatus for measurement of blood pH



Reference

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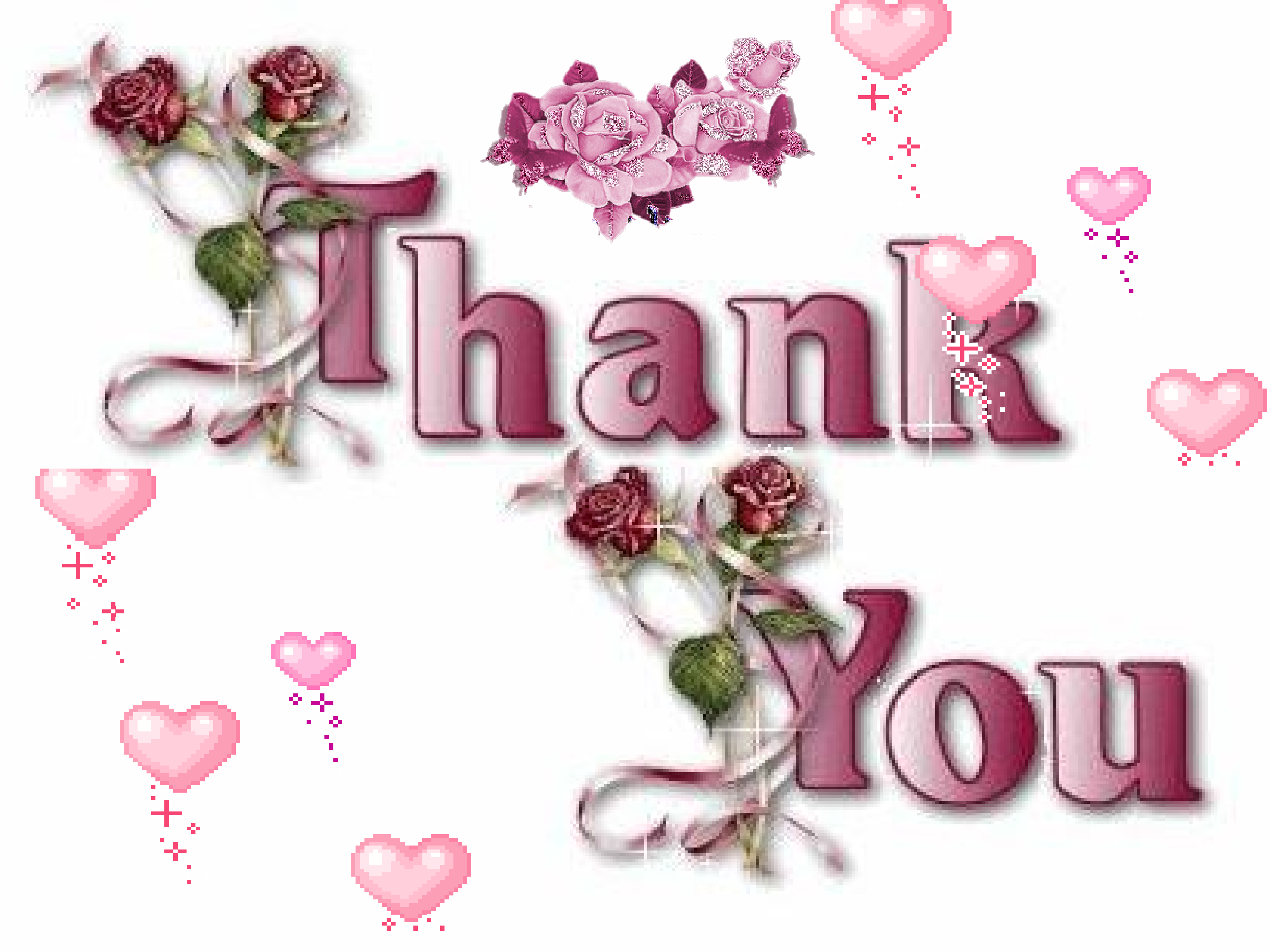
2 Lippincott illustrated reviews integrated systems



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